

6 Flounder in the North Sea, Skagerrak, and Kattegat

fle.27.3a4 – *Platichthys flesus* in Subarea 4 and Division 3.a

6.1 General

Flounder (*Platichthys flesus*) in Subarea 4 and Division 3.a was assessed until 2013 in the Working Group on Assessment of New MoU Species (ICES, 2013a). Because only official landings and survey data were available, flounder was defined as a category 3 species according to the ICES guidelines for data limited stocks (ICES, 2012). Biennial advice for flounder is given since 2013 by ICES (ICES, 2013b) based on survey trends. Since 2015, flounder was included in the official data call for the WGNSSK and discard estimates were included into the assessment. During the WGNSSK 2017, methods to determine MSY proxy reference points were tested. Only the Length Based Indicator (LBI) method was accepted at that time and revealed that the North Sea flounder stock was fished at or below F_{MSY} proxy. Catch advice for flounder was prepared for 2018 and 2019 during the WGNSSK 2017 (ICES, 2017a). However, later in 2017 the combined TAC for dab and flounder was removed (EU COM, 2017/595), and North Sea flounder has become a non-target species with no TAC since then. ICES has not been requested to provide advice on fishing opportunities for flounder for the years 2020 and 2021. The assessment for flounder in Subarea 4 and Division 3.a was benchmarked in 2018 and a SPiCT model (Pedersen and Berg, 2017) was set up to evaluate the stock status of flounder relative to MSY proxies (ICES, 2018). However, updating the SPiCT assessment model with new available data, increased the uncertainties to unacceptable levels. Therefore, the LBI method was used again instead, as it was done for the previous advice (ICES, 2017a). In 2021, precautionary catch advice was again requested for the flounder North Sea stock. Therefore, catch data, survey indices, and the LBI method were updated and presented during the WGNSSK 2021 meeting (ICES, 2021). The empirical *chr* method developed by the WKLIFE X (ICES, 2022) was tested for the flounder stock during the WGNSSK 2023, and catch advice for the years 2024 and 2025 was prepared for the flounder stock applying this method. No advice had to be prepared during the WGNSSK 2024 (ICES, 2024). During the WGNSSK 2025 the assessment was updated and a new catch advice was prepared applying the *chr* method (ICES, 2025): “ICES advises that when the MSY approach is applied, catches in each of the years 2026 and 2027 should be no more than 931 tonnes. If discard rates do not change from the average of the last three years (2022–2024), this implies landings of no more than 649 tonnes.”

6.1.1 Biology and ecosystem aspects

Flounder is a euryhaline flatfish: the life cycle of each individual usually includes marine, brackish, and freshwater habitats. It has a coastal distribution in the Northeast Atlantic, ranging from the White Sea and the Baltic in the north, to the Mediterranean and Black Sea in the south. Flounder can live in low salinity water but they reproduce in water of higher salinity.

Flounder feeds on a wide variety of small invertebrates (mainly polychaete worms, shellfish, and crustaceans), but locally the diet may include small fish species like smelt and gobies. The most intensive feeding occurs in the summer, while food is sparse in the winter.

In the North Sea, Skagerrak and Kattegat flounder spawn between February and April. The adults move further offshore to the 25–40 m deep spawning grounds, the most important of which are situated along the coasts of Belgium, the Netherlands, Germany, and Denmark.

During autumn, both mature and immature flounder withdraw from the inshore and estuarine feeding areas. Juvenile flounder migrate into coastal areas, where they spend the winter.

6.1.2 Stock ID and possible assessment areas

There is no information about stock identity and possible stock assessment areas in the North Sea, Skagerrak and Kattegat. Within the North Sea there may exist a number of sub-populations (ICES 2013a). The distributions derived by IBTS data show that there might be limited connectivity between the North Sea and the Kattegat area, and at least two separate stocks might exist. However, the latter would require a thorough analysis of biological data and possibly genetic methods. This was discussed during the WGNSSK 2024 in a joint session with the WGBFAS 2024.

6.1.3 Management regulations

Flounder is mainly a bycatch species in fisheries for plaice and sole. The discard rates for flounder can be very high in certain métiers and overall it ranged between 12% and 49% in the period for which discard data are available (2002–2024). No minimum landing size is defined for flounder. Prior to 2017, a TAC was set in EU waters of Division 2a and Subarea 4 together with dab (*Limanda limanda*). This combined TAC was never fully utilized. In 2017, the European Commission requested ICES to evaluate the possible effects on the stocks of flounder and dab having no TAC. ICES advised that given the current fishing patterns of the main fleets catching flounder and dab, which are the same fleets targeting plaice and sole, the risk of having no TAC for the flounder and dab stock is considered to be low (ICES, 2017b). However, if this situation changes or flounder is no longer within safe biological limits, the advice to remove the TAC on flounder would need to be reconsidered. The European Commission removed the combined TAC for these two stocks in 2017 (EU COM, 2017/595).

6.2 Fisheries data

6.2.1 Historical landings

In the North Sea and in the Skagerrak and Kattegat, flounder is mainly a bycatch in the fishery for commercially more important flatfish, such as sole and plaice, and in the mixed demersal fisheries. The largest part of total official landings was reported for the North Sea (Subarea 4), while the official landings in the Skagerrak and Kattegat (Division 3.a) decreased considerably, especially for the last decade (Figure 6.1; Table 6.6). Nearly all of the flounder landings in Division 3.a were taken by Denmark (Figure 6.2). Only Sweden landed a small amount of flounder in recent years. The overall trend in landings in Division 3.a is decreasing since the mid of the eighties. The apparent decrease in official landings between 1984 and 1997 in the North Sea (Figure 6.3; Table 6.4) is due to unreported landings by the Netherlands. Further, there seems to be an issue with Danish and German official landings in Subarea 4 which drastically dropped after 1997 (Figure 6.3, red and black bars). At least the drastic decline in Danish landings could be explained by a combined TAC for dab and flounder, which was established in 1998, i.e. that before 1998 partly combined dab and flounder landings may have been reported by the Danish fishery. Another reason could be misreporting to flounder from other quota species from the fishery in area 4 before the TAC came into force in 1998.

Since 1950, annual landings from the North Sea have fluctuated, without any clear pattern (Figure 6.1). During the last decade, landings declined considerably. This decline goes hand in hand with a reduction in fishing effort of bottom trawl fleets in the North Sea since 2000 (STECF, 2016). The lowest official landings were reported for 2023 with 727 tonnes, compared to 820 tonnes in

2024. The decreasing trend in landings continued, and so the 2024 and 2023 landings were the lowest reported landings in the whole time series since 1950. In Division 3.a, annual landings, in general, have decreased sharply from the mid-1980s until 2015 and are on a very low level since then (Figure 6.2). The observed landings for 2024 are 96 tonnes, which is among the lowest values of the whole time series.

Flounder is of relatively little commercial importance in the North Sea and the Skagerrak/Kattegat. Landings data may have been misreported in previous years. However, the amount of misreporting is not known. In addition, the official landings may not reflect the total catches, because flounder is often discarded and discarding is influenced by the prices and the availability of other, commercially more important species. Therefore, the total catch cannot be estimated for years without observations on discards.

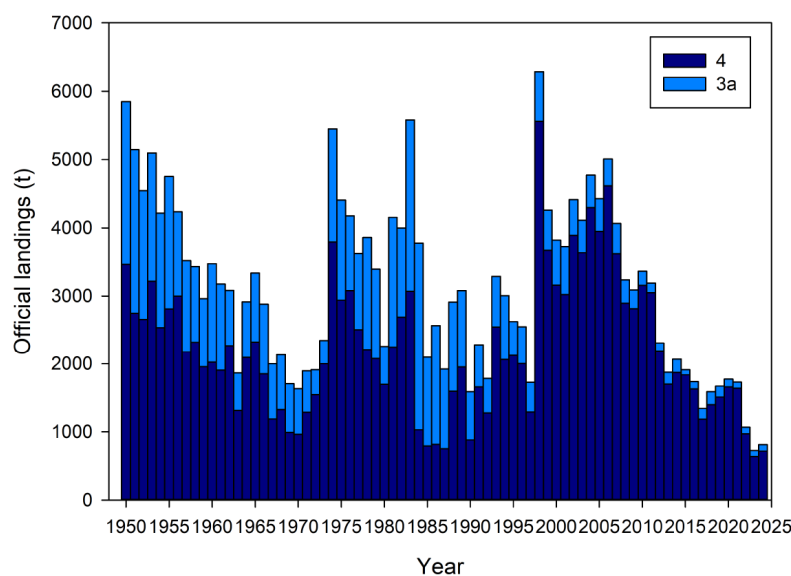


Figure 6.1. Flounder in Subarea 4 and Division 3.a. Official landings in tonnes of flounder by area 1950–2024.

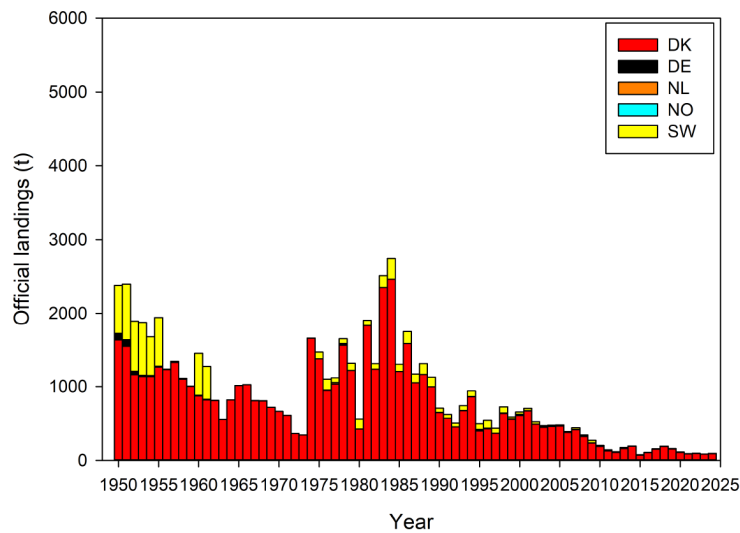


Figure 6.2. Flounder in Subarea 4 and Division 3.a. Official landings in tonnes of flounder in ICES Division 3.a by country. 1950–2024.

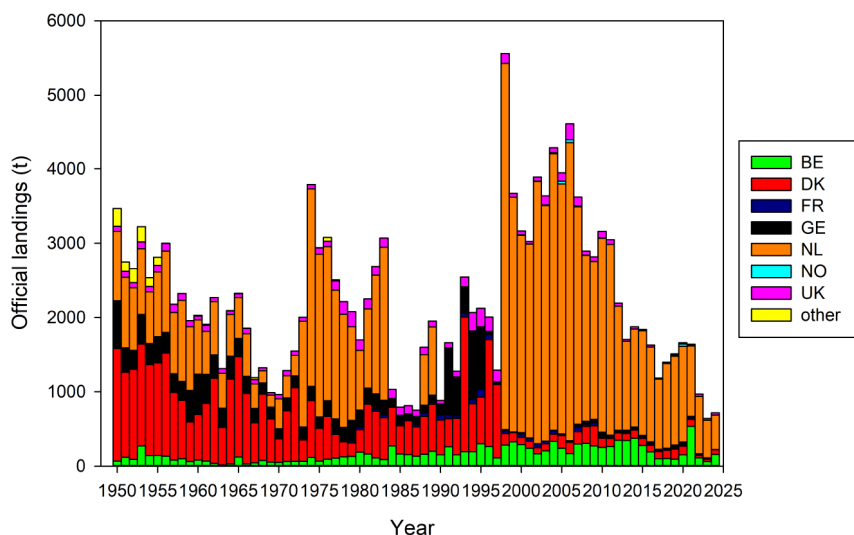


Figure 6.3. Flounder in Subarea 4 and Division 3.a. Official landings of flounder in ICES Subarea 4 by country 1950–2024.

6.2.2 InterCatch

Flounder landings and discards data from 2002–2024 were available in the InterCatch system for the current assessment year (ICES WGNSSK 2025). In 2025, UK England updated its 2021 to 2023 data, which resulted in no changes for the years 2021 and 2022, but one tonne more landings and 16 tonnes more discards for the year 2023.

In general, it was tried only to raise equivalent or similar métiers with each other in InterCatch. Discard information was provided for 60% of all métiers in 2024 (Figure 6.4). However, for a number of métiers zero landings were reported, but no further discard information. For these métiers no raising with InterCatch was possible and it remains unclear if discards occurred, although it seems likely. A further problem in the estimation of total flounder discards may be the TBB_CRU_16-32_0_0_all fleets, targeting brown shrimp in coastal areas of the South-eastern North Sea which is the main distribution area of flounder in the North Sea. Only few observations are available, e.g. Germany only reports discards for the shrimper fleet 2nd quarter with 66.346 t discards and 0.028 t landings, resulting in a discard ratio of 2370.143. Given the low

sampling effort, this value seems highly uncertain and not suitable to raise other fleets. In contrast, discards reported for the Danish shrimp fleet are much lower ranging from 0 to 0.023 t, with zero landings. And the Dutch and Belgium fleets report quite high landings (100 tonnes in total) but provided no discard information.

In 2024, by far the largest proportion of landings (369 tonnes, ~46% of total landings) was reported by Dutch beam trawlers (TBB_DEF_70-99_0_0_all), followed by the Belgian TBB_DEF_70-99_0_0_all fleet (114 tonnes, ~14%; Figure 6.5). In addition, the Dutch and Belgian shrimp fleets (TBB_CRU_16-31_0_0_all) landed considerable amounts of flounder: 100 tonnes in total. The highest discards in 2024 were reported for the Danish OTB_CRU_90-119_0_0_all_FDF métier (~80 tonnes), followed by the German shrimp fleet TBB_CRU_16-32_0_0_all métier with ~66 tonnes, and the Dutch TBB_DEF_70-99_0_0_all métier with ~19 tonnes.

The largest number of raised discards were estimated for the Danish gillnetter fleet (GNS_DEF_120-219_0_0_all; 208 tonnes), similar as for the assessments in previous years. However, it has to be noted here, that only Swedish passive gears from Division 3a were available for the raising (Figure 6.6).

The largest total catch estimated in 2024 (Figure 6.7) was taken by Denmark (483 tonnes), followed by The Netherlands (466 tonnes), Belgium (168 tonnes) and Germany (77 tonnes). The total catch estimated with InterCatch was 1300 tonnes from which 797 tonnes were landings (compared to 820 tonnes reported official landings) and 503 tonnes discards (39% of the total catch). However, it should be noted that not all métiers were sampled in every quarter and that the raising procedure may not be adequate for all cases.

In general, it was attempted to use the same groupings for discard raising as for the previous data years. However, this was not possible for all cases and compared to the previous year slight changes had to be made. The grouping is based on gear type and mesh size over areas and season. For the sample allocation scheme one landing and one discard group was set up, because data availability did not allow for a higher resolution. A separate discard group for the shrimp fleets was set up (TBB_CRU_16-31_0_0_all), because the length distribution with only small individuals (≤ 10 cm) differs largely from the fleets operating more offshore. The following groupings were used for the 2024 data discard raising:

- Group 1: TBB_DEF_70-99_0_0_all raised with all TBB_DEF_70-99_0_0_all métiers.
 - Group 2: MIS_MIS_0_0_0_HC raised with all other métiers because no specific MIS_MIS_0_0_0_HC data were available.
 - Group 3: all OTB, SSC, SDN, 70 – 119 métiers raised with all other métiers of same type and mesh size.
 - Group 4: All passive gears raised with all passive gears (only SWE métiers from 3a available).
 - Group 5: OTB_DEF ≥ 120 with all OTB_DEF ≥ 120 métiers
 - Group 6: SDN_SSC_DEF ≥ 120 with all other SDN_SSC_DEF ≥ 120 (only DEN métier with zero discards available)
 - Group 7: TBB_DEF $\geq 100_0_0_0$ raised with all TBB_DEF_70-99_0_0_all métiers
- Not raised: MIS_MIS_0_0_0_IBC, OTB_SPF_32-69_0_0_all, TBB_CRU_16-32, DRB_all_0_0_all

Length allocations for 2024 data: one discard group (including BMS L and LogBook D, excluding TBB_CRU_16-31_0_0_all data) and one landing group. In addition, one separate group for TBB_CRU_16-31_0_0_all discards was set up.

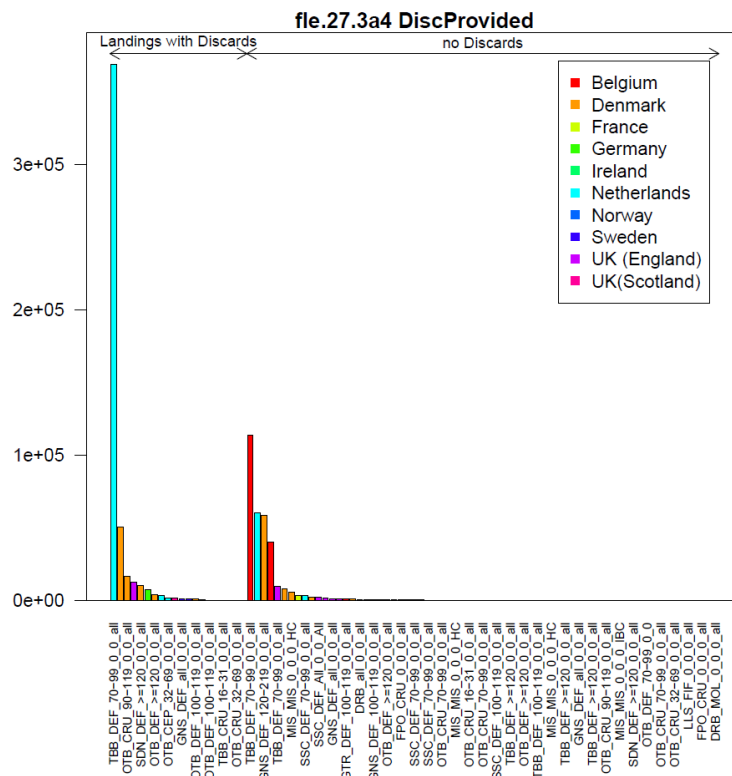


Figure 6.4. Flounder in Subarea 4 and Division 3.a. Provision of discards information by country and fleets imported to InterCatch for 2024 data.

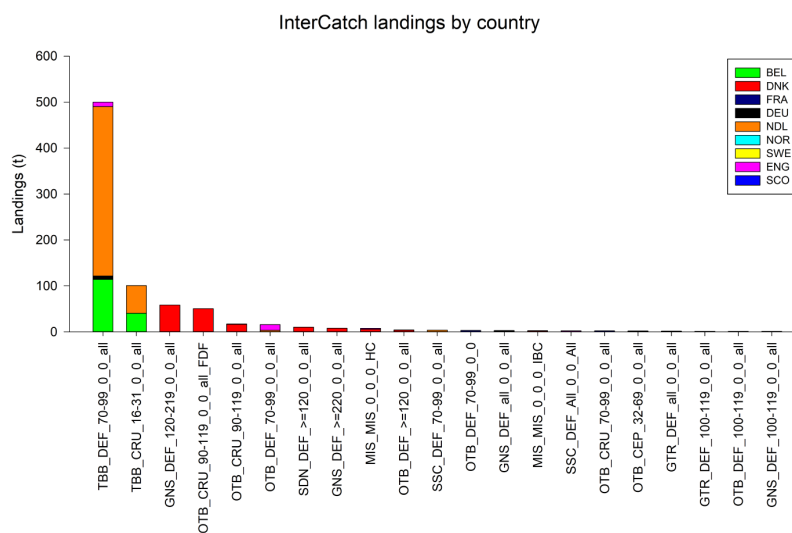


Figure 6.5. Flounder in Subarea 4 and Division 3.a. Flounder landings by métier and country in 2024 imported to InterCatch.

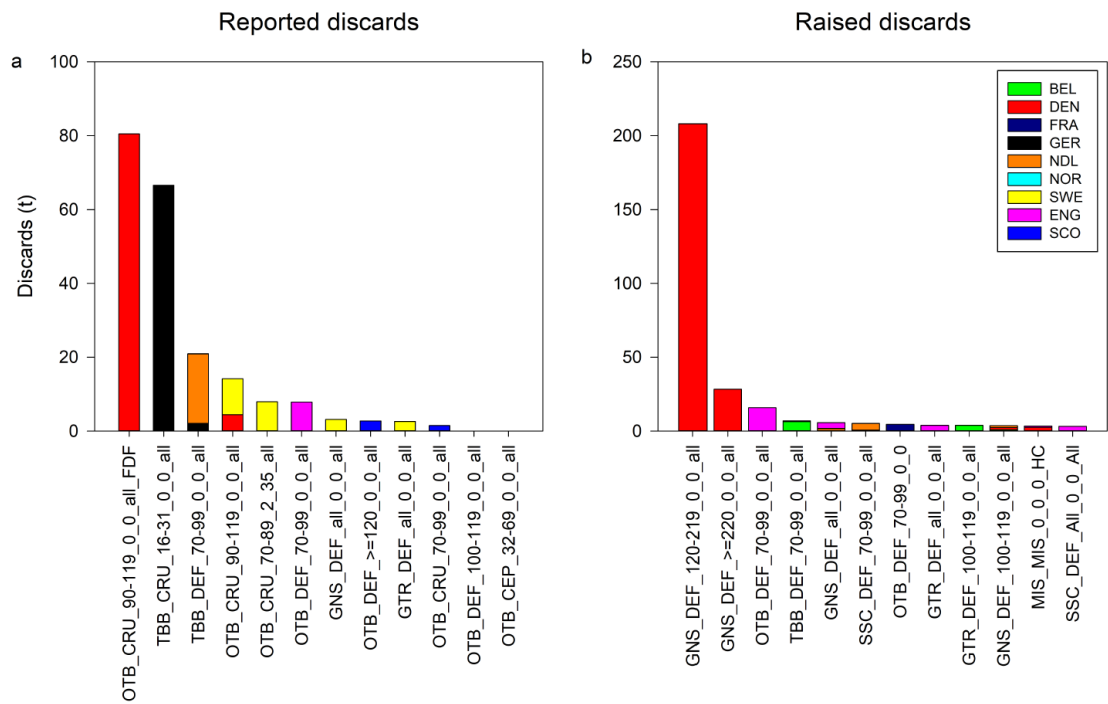


Figure 6.6. Flounder in Subarea 4 and Division 3.a. Flounder discards by métier and country in 2024. Reported discards panel (a), raised discards panel (b).

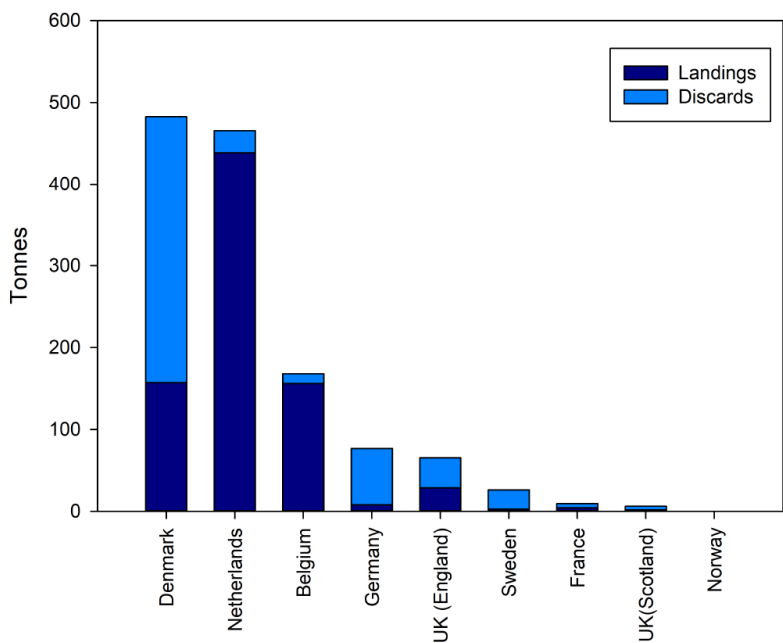


Figure 6.7. Flounder in Subarea 4 and Division 3.a. Flounder landings and discards by country in 2024 estimated with InterCatch.

6.2.3 Survey data/recruit series

Several surveys in the North Sea, Skagerrak and Kattegat provide information on distribution, abundance and length composition of flounder. The most relevant survey for flounder is probably the North Sea International Bottom Trawl Survey in quarter 1 (NS-IBTSQ1) because it covers

the whole distribution area of the stock and shows an even higher catchability compared to the beam trawl surveys conducted in quarter 3 (BTS). However, the NS-IBTSQ1 uses a bottom trawl, which is not necessarily designed to catch flatfish. Further, it should be noted here that the NS-IBTSQ1 was not fully standardized before 1983. Therefore, index data before this year should be interpreted with caution and are not presented in this report. The beam trawl surveys (BTS) use a beam trawl and are designed for catching flatfish. However, they are carried out in quarter 3, during which flounder is distributed in more coastal, shallow and brackish waters in the river estuaries and the Wadden Sea areas (Figure 6.8). Biological data available from the NS-IBTSQ1 survey is displayed in Figure 6.9. and Figure 6.10.

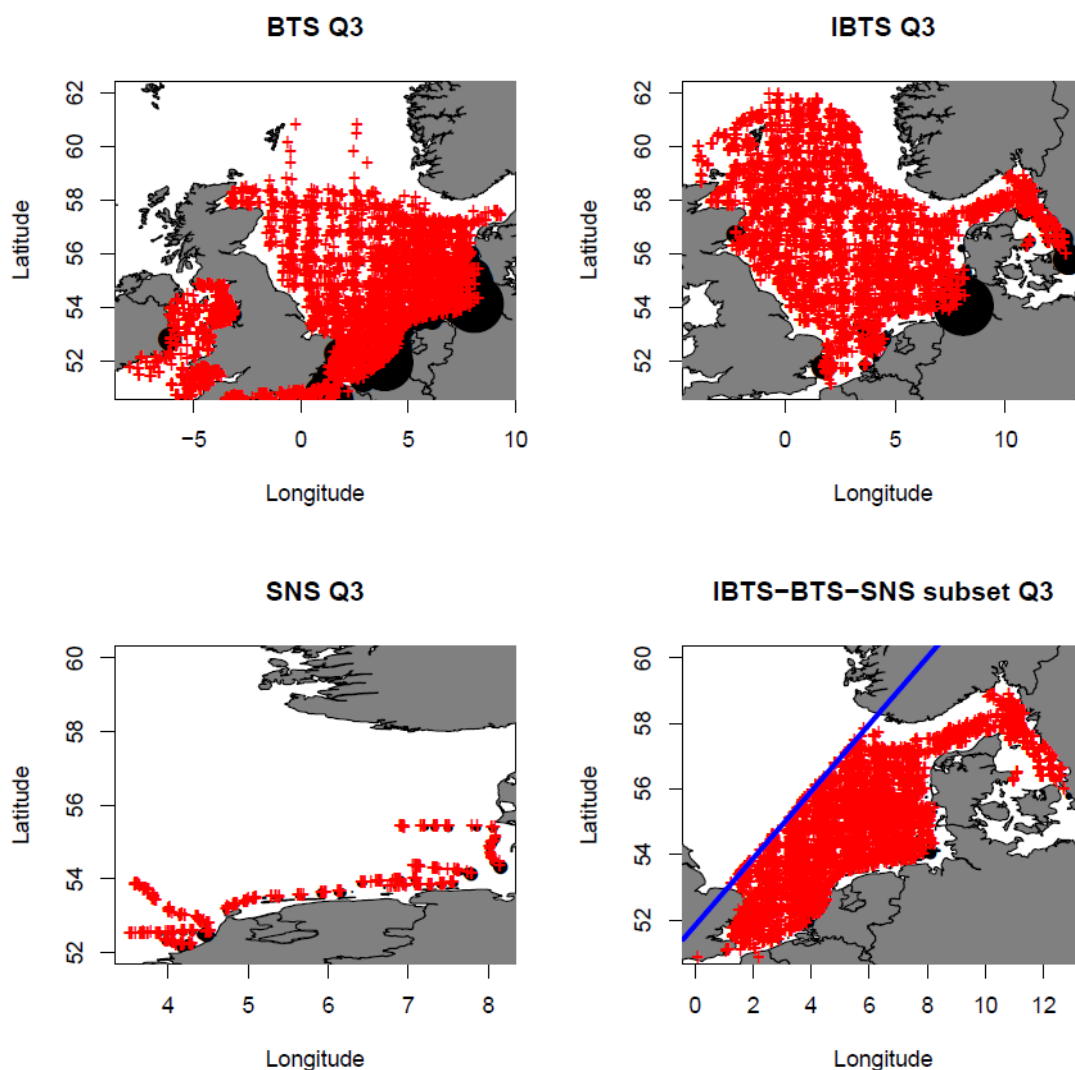


Figure 6.8. Flounder in Subarea 4 and Division 3.a. Distribution of flounder derived from different trawl surveys in Subarea 4 and Division 3.a and the defined index area (lower right panel).

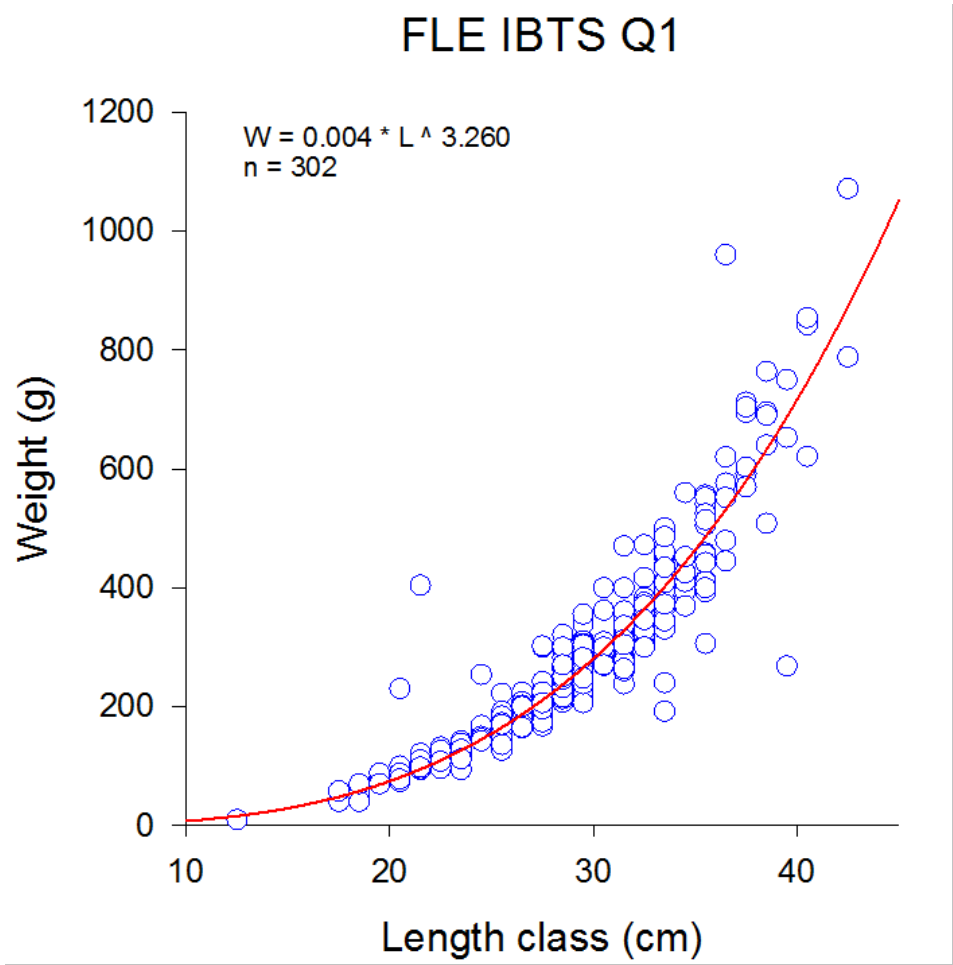


Figure 6.9. Flounder in Subarea 4 and Division 3.a. Length weight relationship of flounder derived from NS-IBTSQ1 data.

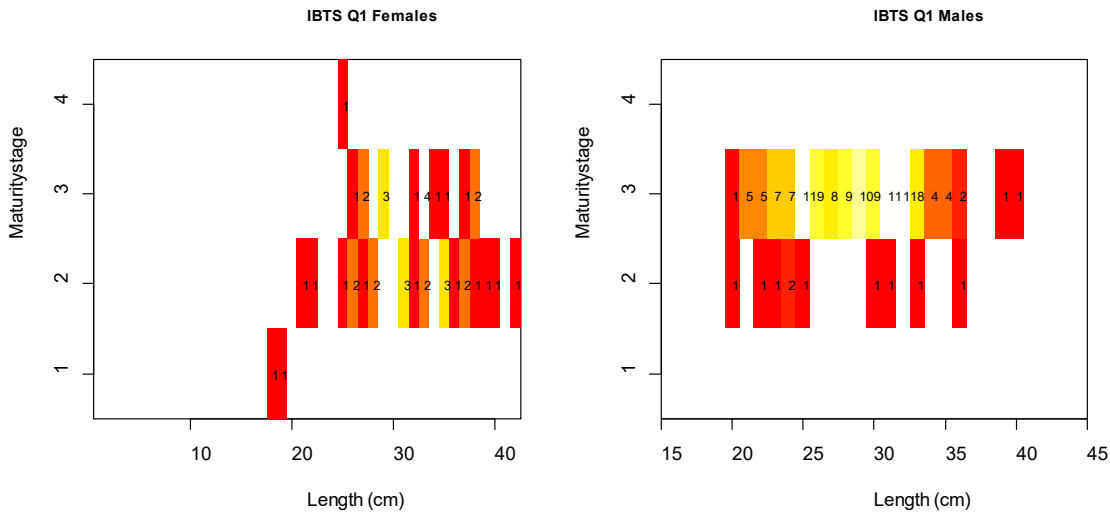


Figure 6.10. Flounder in Subarea 4 and Division 3.a. Maturity at length of female and male flounder derived from IBTS–Q1 data.

Survey indices

The flounder assessment was benchmarked in 2018 and two survey indices were constructed: a NS-IBTS Q1 and a combined quarter 3 index (combining NS-IBTS Q3, BTS Q3, SNS Q3). Both indices were modelled with the deltaGAM method (Berg *et al.*, 2014). For both indices, the index area was defined, based on the species distribution from the hauls (Figure 6.8 lower right panel) which is restricted to the south-eastern part of the North Sea and Division 3.a. In quarter 3, four gear types were used in the different beam trawl surveys (BT8, BT7, BT6, and BT4) and the GOV in the NS-IBTS survey. Therefore, a gear effect was included in order to model a combined quarter 3 index for flounder (equation 2 below).

The following models were formulated:

Quarter 1

$$g(\mu_i) = Year(i) + f_1(lon_i, lat_i) + f_2(depth_i) + \log(HaulDur_i) \quad (1)$$

Quarter 3 – with gear effect

$$g(\mu_i) = Year(i) + Gear(i) + f_1(lon_i + lat_i) + f_2(depth_i) + \log(HaulDur_i) \quad (2)$$

The NS-IBTSQ1 index shows higher values at the beginning of the time series, but decreases towards 2000 (Figure 6.11 blue line; Table 6.8). Since 2000, the index increased again until 2008. Since then, the index decreased, with the lowest observed value in 2023. The 2024 observation is again higher, but still on a comparative low level. The combined quarter 3 index (Figure 6.11 red line) in general follows the trend of the NS-IBTSQ1 index. A sharp drop of the combined Q3 index value was observed in 2021, resulting in the lowest observed value for the whole time series. The overall trend of both indices is similar with higher observed values at the beginning of the time series and an overall decreasing trend from 2008 onwards.

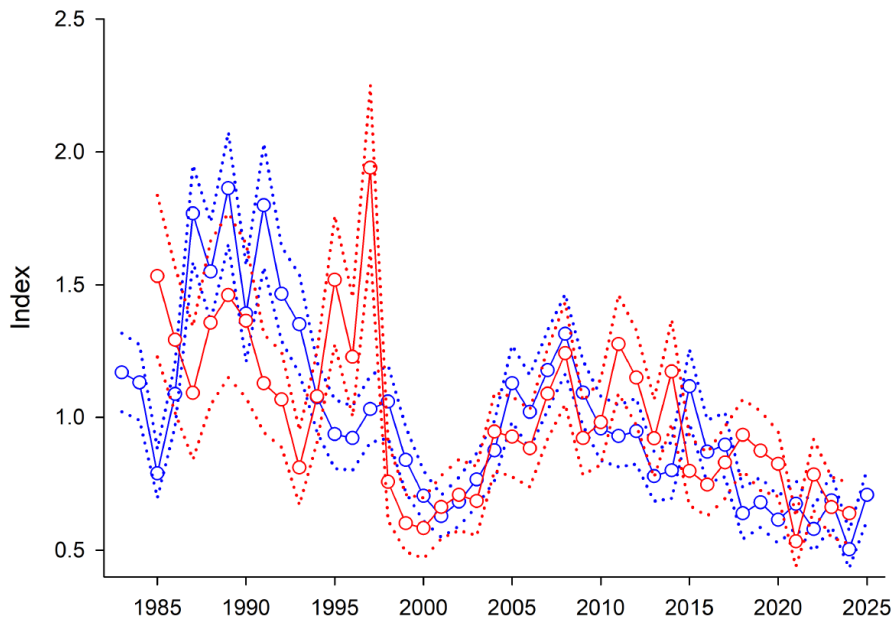


Figure 6.11. Flounder in Subarea 4 and Division 3.a. IBTS Quarter 1 biomass index (blue line) and combined quarter 3 biomass index (red line). Indices standardized to their means, dotted lines display sd.

6.3 MSY proxy analyses for flounder in Subarea 4 and Division 3.a.

6.3.1 Length based indicators

Flounder length samples (sex combined) from commercial catches were provided in InterCatch format for the years 2014–2024. These data were used for the analyses of MSY proxies applying the Length Based Indicator method (LBI; ICES, 2017a). The commercial length data show incoming recruitment peaks for some of the years (Figure 6.12). Since the LBI method assumes constant recruitment, the data sets were reduced by length classes below 16 cm (corresponding to ages below 2 years) for the analyses. Further, the length distributions were binned to 20 mm length classes. The method also requires growth parameters, which were taken either from literature (Froese and Sampang, 2013; Table 6.1) or estimated based on the available survey or InterCatch data. The L_{inf} was recalculated this year using InterCatch length distribution and the empirical formula by Garcia-Carreras *et al.* (2016):

$$\log_{10}L_{\infty} = 0.068260 (\pm 0.010451) + 0.969112 (\pm 0.006318) \log_{10}L_{max},$$

where L_{max} is defined as the 99% percentile of the commercial length distribution (39.5 cm; Figure 6.13). This resulted in the applied L_{inf} of 41.3 cm. The estimated L_{inf} value did not change, taking into account 2024 data.

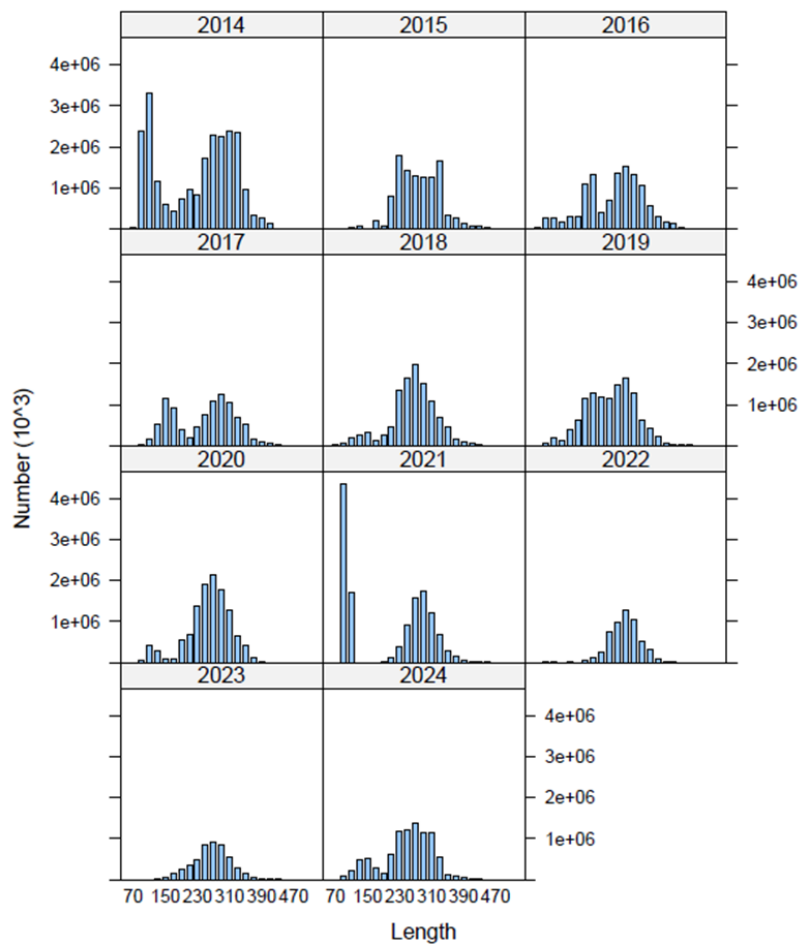


Figure 6.12. Flounder in Subarea 4 and Division 3.a. Length distributions (20 mm length classes) from InterCatch 2014–2024. Length <=16 cm were removed from the subsequent analysis to account for incoming recruitment.

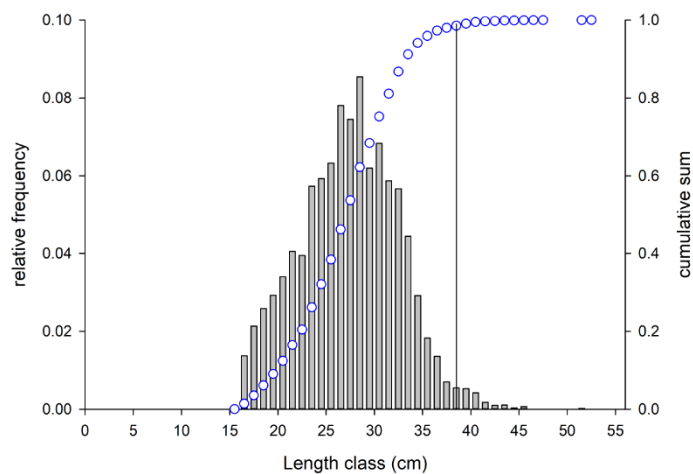


Figure 6.13. Flounder in Subarea 4 and Division 3.a. InterCatch relative length distribution (2014–2024) with the cumulative sum. Vertical line displays the 99% percentile of the distribution (39.5 cm).

The results of the LBI method showed that all of the indicators are above the reference points for 2024 (Table 6.2), except the P_{mega} indicator which was at 25%. The L_c/L_{mat} ratio fluctuated around 1 between 2014 and 2019, but was above 1 since 2020 (Table 6.2; Figure 6.14). In terms of the F_{MSY}

proxy, the indicator ratio ($L_{mean}/L_{F=M}$) is above 1 for all the years (Table 6.2; Figure 6.16). From these results it was concluded that flounder is currently exploited below F_{MSY} .

Table 6.1. Flounder in Subarea 4 and Division 3.a. Parameters used as input for the LBI method.

Parameter	Sex combined
von Bertalanffy L_{∞} (cm)	41.3
von Bertalanffy k (yr^{-1})	0.36
Length-weight a	0.004
Length weight b	3.260
Natural mortality M (yr^{-1})	0.2
Length-at-maturity (cm)	21
Natural mortality M	0.2

Table 6.2. Flounder in Subarea 4 and Division 3.a. Length Based Indicator table displaying the reference points and indicators based in InterCatch length sample data 2014–2024.

Ref	Conservation				Optimizing Yield	MSY
	LC/L_{mat}	$L_{25\%}/L_{mat}$	$L_{max5\%}/L_{inf}$	P_{mega}	L_{mean}/L_{opt}	$L_{mean}/L_{F=M}$
	>1	>1	>0.8	>30%	~1(>0.9)	≥1
2014	0.90	1.21	0.93	0.42	1.05	1.18
2015	1.10	1.12	0.94	0.36	1.05	1.05
2016	0.90	1.02	0.96	0.35	1.01	1.13
2017	0.81	1.17	0.93	0.37	1.02	1.22
2018	1.10	1.17	0.91	0.26	1.03	1.03
2019	0.90	1.02	0.89	0.24	0.98	1.10
2020	1.10	1.17	0.87	0.23	1.02	1.02
2021	1.19	1.26	0.90	0.34	1.08	1.02
2022	1.19	1.26	0.88	0.37	1.08	1.02
2023	1.10	1.17	0.87	0.22	1.03	1.03
2024	1.10	1.12	0.86	0.25	1.02	1.02

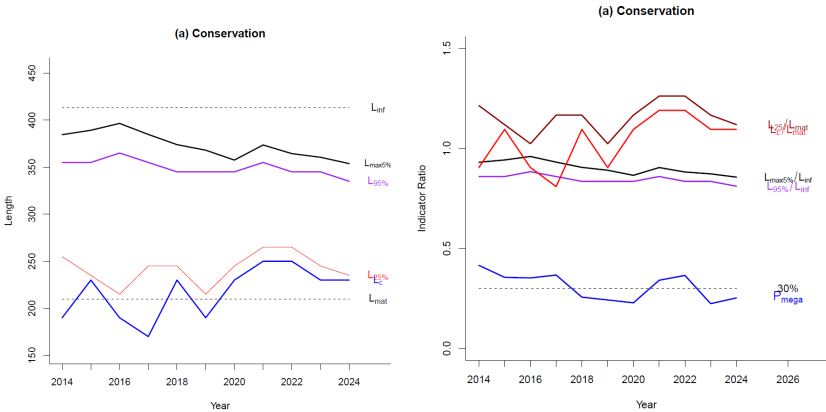


Figure 6.14. Flounder in Subarea 4 and Division 3.a. Conservation indicators (left panel, length in mm) and indicator ratios (right panel).

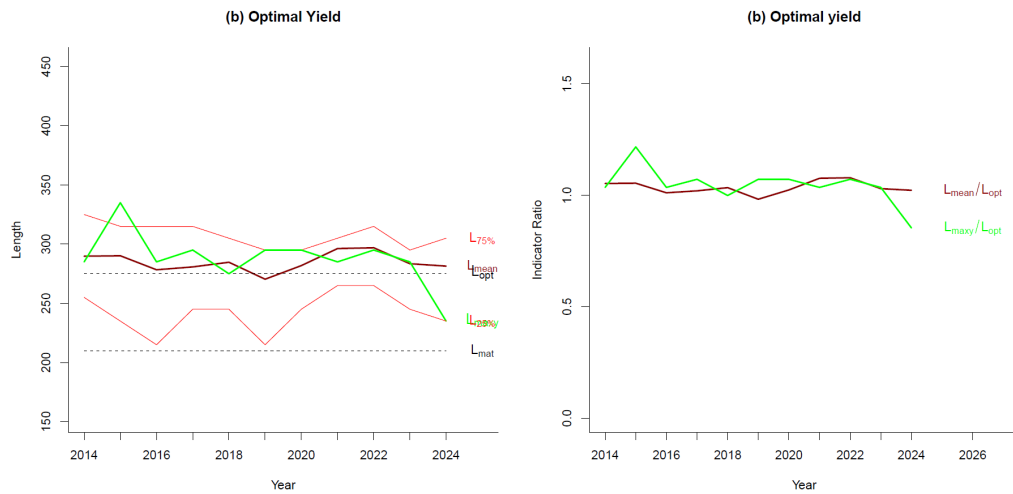


Figure 6.15. Flounder in Subarea 4 and Division 3.a. Optimum yield indicators (left panel, length in mm) and indicator ratios (right panel).

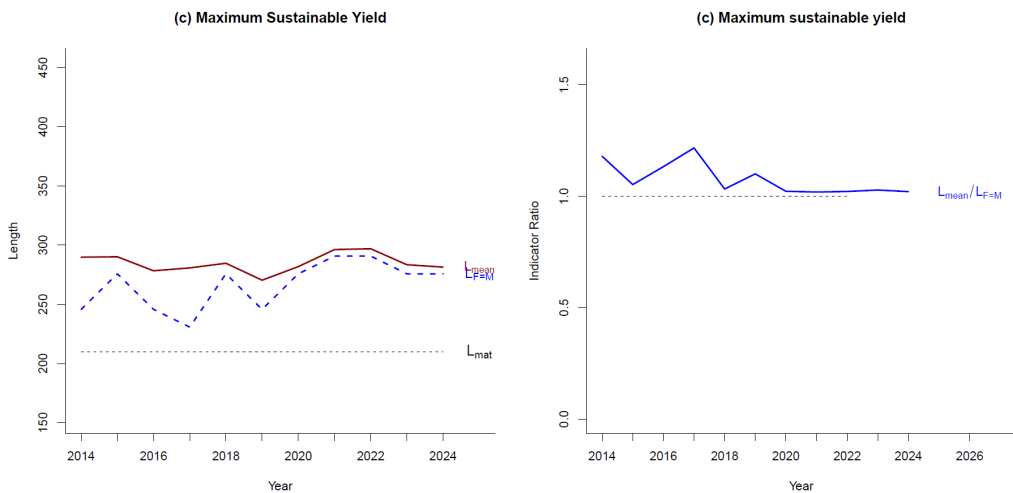


Figure 6.16. Flounder in Subarea 4 and Division 3.a. Maximum sustainable yield indicator (left panel, length in mm) and indicator ratio (right panel).

6.4 Empirical *chr* rule

The constant harvest rate (*chr*) rule was applied in the case of flounder, because the available growth parameter $k = 0.36$ (Table 6.1).

The *chr* rule applies a constant harvest rate that is considered a proxy for an MSY harvest rate, which is then applied to a biomass index:

$$C_{y+1} = I_{y-1} \times F_{MSYproxy} \times b \times m$$

with C_{y+1} is the catch advice, I_{y-1} the most recent index value, $F_{MSYproxy}$ is the ratio C_y / I_y for which the quantity $f > 1$, where f is the ratio of the mean length L_m in the catch and $L_{F=M}$ (Figure 6.17; inverse f), b is a biomass safeguard, and m is a multiplier applied to the *chr* rule to maintain the probability of the biomass declining below B_{lim} to less than 5%.

The index used as the basis for the *chr* rule calculation is the NS-IBTS Q1 (Figure 6.17). The biomass safeguard b is determined by the ratio between the most recent index value to $I_{trigger}$ (with $I_{trigger} = 1.4 \times I_{loss}$; I_{loss} = defined as lowest observed index value = the index value for 2022). When the most recent index value is greater than $I_{trigger}$, b is set to 1. However, for flounder this was not the case, and b is calculated to be 0.87 (Table 6.3). The precautionary multiplier m is applied with the *chr* rule to maintain the probability of the biomass declining below B_{lim} to less than 5%. Following the guidelines (ICES 2022), this precautionary multiplier should be set to 0.5 when applying the *chr* rule. An uncertainty cap (*stability clause*) could be applied to limit the amount the advised catch can change upwards or downwards between years (+20% / -30%). This uncertainty cap was not applied. Based on the latest catch advice (864 tonnes), the *chr* rule resulted in a catch advice of 931 tonnes, which corresponds to 649 tonnes landings, given the average discard rate of the most recent three years (30%). The change in advice is due to an increased biomass index. The summary of the *chr* rule for flounder is displayed in table 6.3 and all input data are listed in table 6.8.

Table 6.3 Flounder in Subarea 4 and Division 3.a. The basis for the catch scenarios. Catches are in tonnes.*

Previous catch advice A_y (advised catch for 2024 and 2025)	864 tonnes
Biomass index	
I: most recent biomass index (I_{2025})	19.590
MSY proxy harvest rate	
HR_{MSY} proxy: MSY proxy harvest rate (average of the ratio of catch to biomass index for the years for which $f > 1$, where $f = L_{mean}/L_{F=M}$)	108.932
Biomass safeguard	
$I_{loss} = I_{2022}$	16.032
Index trigger value ($I_{trigger} = I_{loss} \times 1.4$)	22.445
b : index relative to trigger value, $\min\{I_{2025}/I_{trigger}, 1\}$	0.87
Precautionary multiplier to maintain biomass above B_{lim} with 95% probability	
m : multiplier (generic multiplier based on life history)	0.5
CHR calculation**	931 tonnes
Stability clause (+20%/-30% compared to A_y , only considered if $b=1$)	Not applied
Discard rate	30%
Catch advice A_{y+1} for 2026 and 2027**	931 tonnes
Landings corresponding to the advice for 2024 and 2025***	649 tonnes
% advice change^	8 %

* The figures in the table are rounded. Calculations were done with unrounded inputs, and computed values may not match exactly when calculated using the rounded figures in the table.

** Formula: $A_{y+1} = I \times F_{MSY} \text{ proxy} \times b \times m$, limited by stability clause if applicable.

*** $A_{y+1} \times (1 - \text{Discard rate})$

^ Advice value for 2026 and 2027 relative to the advice value for 2024 and 2025 (1650 tonnes).

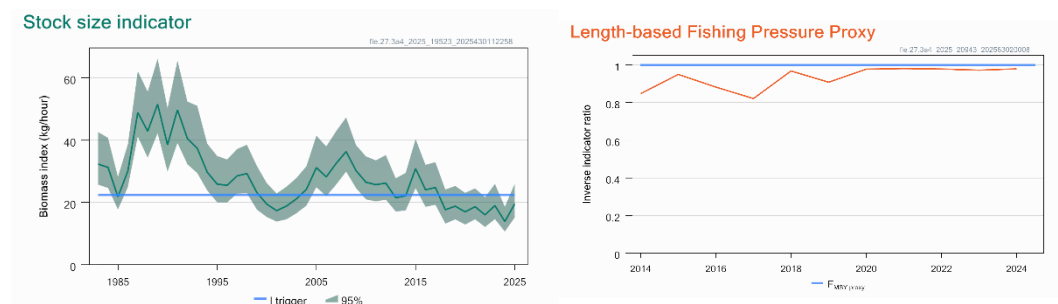


Figure 6.17. Flounder in Subarea 4 and Division 3.a. Left panel: Stock size indicator (biomass index) from the North Sea International Bottom Trawl Survey Q1 (NS-IBTS); right panel: Indicator ratio $LF = M/L_{mean}$ (inverse of the indicator ratio, f) from the length-based indicator (LBI) method is used for the evaluation of the exploitation status. The proxy fishing

pressure is less than that corresponding to the FMSY proxy ($LF = M$) when the indicator ratio value is lower than 1 (shown by the horizontal orange line).

6.5 Issues List

- Métiers with zero landings and no reported discards from which discarding is suspected: No raising possible for these cases. What is the possible impact on catch estimation? Are there other ways to estimate discards for these métiers?
- No suitable data available for the shrimper fleets operating in coastal waters. Raising highly uncertain for these fleets. What is the possible impact on catch estimation? Is there another way to estimate the discards of these fleets?
- Investigate what could be done/changed to improve the SPiCT model.
- Investigate the use of alternative stock indices (DYFS, others?) which are able to better reflect the stock status.
- Investigate available biological data, evaluate growth parameter k .
- Evaluate the use of a time-varying L_c (mean length in the catch) in the LBI method.
- It should be investigated if the stocks in the North Sea and Kattegat are two separate stock units which is indicated by the survey distribution of the NS-IBTSQ1.

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6.7 Tables

Table 6.4. Flounder in Subarea 4 and Division 3.a. Flounder official landings by country in ICES Subarea 4.

Year	Belgium	Denmark	France	Germany	Netherlands	Norway	UK	Other	Total
1950	67	1514	0	641	937	0	67	241	3467
1951	119	1143	0	329	949	0	81	127	2748
1952	91	1210	0	257	841	0	71	186	2656
1953	270	1372	0	397	886	0	92	203	3220
1954	142	1225	0	281	696	0	71	121	2536
1955	145	1244	0	353	871	0	88	109	2810
1956	132	1389	0	277	1097	0	102	2	2999
1957	81	910	0	250	825	0	112	0	2178
1958	99	784	0	257	1088	0	94	0	2322
1959	62	533	0	424	857	0	79	1	1956
1960	82	614	0	540	733	0	49	8	2026
1961	68	776	0	390	579	0	81	13	1907
1962	37	1146	0	313	717	0	53	2	2268
1963	16	501	0	263	467	0	65	0	1312
1964	30	1141	0	305	563	0	48	6	2093
1965	121	1349	0	248	549	0	54	3	2324
1966	32	946	0	229	573	0	71	2	1853
1967	43	540	0	193	331	0	57	25	1189
1968	75	894	0	152	160	0	43	1	1325
1969	54	582	0	158	161	0	33	0	988

Year	Belgium	Denmark	France	Germany	Netherlands	Norway	UK	Other	Total
1970	50	316	0	135	405	0	57	0	963
1971	60	685	0	173	297	0	70	0	1285
1972	63	991	0	159	275	0	60	0	1548
1973	63	290	0	172	1424	0	53	0	2002
1974	115	766	0	190	2661	0	58	0	3790
1975	68	437	0	155	2191	0	87	1	2939
1976	94	575	0	209	2077	0	70	54	3079
1977	107	320	0	208	1732	0	127	11	2505
1978	122	203	0	198	1519	0	169	0	2211
1979	129	181	31	275	1260	0	201	0	2077
1980	190	300	33	229	806	0	140	0	1698
1981	164	669	14	200	1068	0	133	0	2248
1982	110	630	31	200	1597	0	121	0	2689
1983	88	564	36	197	2059	0	125	0	3069
1984	272	518	15	103	0	0	122	0	1030
1985	163	379	14	128	0	0	109	0	793
1986	155	456	1	91	0	0	111	0	814
1987	132	394	32	106	0	0	90	0	754
1988	160	509	44	105	682	0	98	0	1598
1989	200	632	28	95	916	0	80	0	1951
1990	153	467	69	147	0	0	45	0	881
1991	260	377	51	902	0	0	69	0	1659
1992	152	492	35	521	0	0	76	0	1276
1993	194	1812	47	356	0	0	136	0	2545
1994	196	642	57	921	0	0	247	0	2063
1995	301	628	103	843	0	0	250	0	2125
1996	262	1439	68	43	0	0	193	0	2005
1997	110	988	10	25	0	0	157	0	1290
1998	283	154	40	13	4938	0	132	0	5560

Year	Belgium	Denmark	France	Germany	Netherlands	Norway	UK	Other	Total
1999	326	123	0	11	3158	0	54	0	3672
2000	289	100	46	17	2656	5	52	0	3165
2001	241	92	42	4	2608	3	32	0	3022
2002	165	83	51	2	3531	3	55	0	3890
2003	206	94	33	3	3172	9	120	0	3637
2004	335	96	46	5	3720	18	74	0	4294
2005	241	171	17	5	3363	38	111	0	3946
2006	168	152	19	2	4020	39	216	0	4616
2007	298	166	56	45	2925	11	119	0	3620
2008	306	228	30	39	2231	3	57	0	2894
2009	272	273	38	46	2124	3	59	0	2815
2010	251	126	20	58	2612	6	87	0	3160
2011	262	112	17	25	2566	1	65	0	3048
2012	348	100	11	23	1672	0	38	0	2192
2013	346	93	13	28	1199	0	24	0	1703
2014	376	107	15	30	1314	0	31	0	1873
2015	277	97	19	19	1409	0	15	0	1836
2016	193	87	20	27	1277	0	25	0	1629
2017	97	101	0	28	944	1	14	0	1185
2018	102	114	0	24	1139	1	18	0	1398
2019	94	133	9	48	1192	19	15	0	1510
2020	154	114	7	48	1287	30	18	0	1658
2021	134	93	19	21	947	2	19	1	1236
2022	109	35	6	17	780	0	30	0	977
2023*	60	32	4	13	508	1	22	0	640
2024*	156	64	4	8	462	0	30	0	724

*Preliminary catch statistics

Table 6.5. Flounder in Subarea 4 and Division 3.a. Flounder official landings by country in ICES Division 3.a.

Year	Denmark	Germany	Netherlands	Norway	Sweden	Total
1950	1632	92	0	0	657	2381
1951	1548	88	0	0	759	2395
1952	1161	48	0	0	683	1892
1953	1135	17	0	0	724	1876
1954	1138	13	0	0	528	1679
1955	1265	11	0	0	667	1943
1956	1229	6	0	0	0	1235
1957	1331	12	0	0	0	1343
1958	1099	12	0	0	0	1111
1959	1003	3	0	0	0	1006
1960	875	10	0	0	566	1451
1961	821	9	0	0	442	1272
1962	812	3	0	0	0	815
1963	554	0	0	0	0	554
1964	822	1	0	0	0	823
1965	1016	0	0	0	0	1016
1966	1027	0	0	0	0	1027
1967	811	3	0	0	0	814
1968	808	2	0	0	0	810
1969	721	0	0	0	0	721
1970	667	0	0	0	0	667
1971	611	1	0	0	0	612
1972	365	0	0	0	0	365
1973	346	0	0	0	0	346
1974	1656	2	0	0	0	1658
1975	1377	1	0	0	89	1467
1976	949	2	4	0	144	1099
1977	1036	0	19	0	64	1119
1978	1560	10	14	0	64	1648

Year	Denmark	Germany	Netherlands	Norway	Sweden	Total
1979	1219	0	0	0	100	1319
1980	426	0	0	0	135	561
1981	1831	0	0	0	74	1905
1982	1236	0	0	0	75	1311
1983	2352	0	0	0	160	2512
1984	2463	0	0	0	283	2746
1985	1203	0	0	0	102	1305
1986	1585	0	0	0	166	1751
1987	1050	0	0	0	119	1169
1988	1164	0	0	0	149	1313
1989	996	0	0	0	133	1129
1990	650	1	0	0	57	708
1991	574	0	0	0	50	624
1992	455	0	0	0	52	507
1993	673	3	0	0	67	743
1994	865	1	0	0	77	943
1995	403	19	0	0	76	498
1996	429	9	0	0	104	542
1997	367	2	0	0	68	437
1998	637	5	0	0	83	725
1999	558	6	0	0	24	588
2000	609	17	0	0	30	656
2001	672	2	0	1	30	705
2002	493	0	0	1	30	524
2003	452	3	0	0	18	473
2004	462	2	0	0	14	478
2005	467	0	0	0	15	482
2006	380	0	0	0	13	393
2007	419	3	1	0	22	445

Year	Denmark	Germany	Netherlands	Norway	Sweden	Total
2008	326	4	0	0	16	346
2009	238	2	0	0	33	273
2010	188	0	0	0	17	205
2011	129	0	0	0	16	145
2012	110	0	0	0	8	118
2013	162	0	0	0	11	173
2014	190	0	0	0	4	194
2015	74	0	0	0	3	77
2016	106	0	0	0	3	109
2017	153	0	0	1	5	159
2018	189	0	0	0	3	192
2019	155	0	2	0	3	160
2020	111	0	0	0	4	115
2021	90	0	0	0	2	92
2022	96	0	0	0	2	98
2023*	85	0	0	0	2	87
2024*	93	0	0	0	3	96

* preliminary catch statistics

Table 6.6. Flounder in Subarea 4 and Division 3.a. Flounder total official landings by ICES areas.

Year	Division 3.a	Subarea 4	Total
1950	2381	3467	5848
1951	2395	2748	5143
1952	1892	2656	4548
1953	1876	3220	5096
1954	1679	2536	4215
1955	1943	2810	4753
1956	1235	2999	4234
1957	1343	2178	3521
1958	1111	2322	3433

Year	Division 3.a	Subarea 4	Total
1959	1006	1956	2962
1960	1451	2026	3477
1961	1272	1907	3179
1962	815	2268	3083
1963	554	1312	1866
1964	823	2093	2916
1965	1016	2324	3340
1966	1027	1853	2880
1967	814	1189	2003
1968	810	1325	2135
1969	721	988	1709
1970	667	963	1630
1971	612	1285	1897
1972	365	1548	1913
1973	346	2002	2348
1974	1658	3790	5448
1975	1467	2939	4406
1976	1099	3079	4178
1977	1119	2505	3624
1978	1648	2211	3859
1979	1319	2077	3396
1980	561	1698	2259
1981	1905	2248	4153
1982	1311	2689	4000
1983	2512	3069	5581
1984	2746	1030	3776
1985	1305	793	2098
1986	1751	814	2565
1987	1169	754	1923

Year	Division 3.a	Subarea 4	Total
1988	1313	1598	2911
1989	1129	1951	3080
1990	708	881	1589
1991	624	1659	2283
1992	507	1276	1783
1993	743	2545	3288
1994	943	2063	3006
1995	498	2125	2623
1996	542	2005	2547
1997	437	1290	1727
1998	725	5560	6285
1999	588	3672	4260
2000	656	3165	3821
2001	705	3022	3727
2002	524	3890	4414
2003	473	3637	4110
2004	478	4294	4772
2005	482	3946	4428
2006	393	4616	5009
2007	445	3620	4065
2008	346	2894	3240
2009	273	2815	3088
2010	205	3160	3365
2011	145	3048	3193
2012	118	2192	2310
2013	173	1703	1876
2014	194	1873	2067
2015	77	1836	1913
2016	109	1629	1738

Year	Division 3.a	Subarea 4	Total
2017	159	1185	1344
2018	192	1398	1590
2019	160	1510	1670
2020	115	1658	1773
2021	92	1236	1328
2022	98	977	1075
2023*	87	640	727
2024*	96	724	820

* preliminary catch statistics

Table 6.7. Flounder in Subarea 4 and Division 3.a. Total official landings, InterCatch landings, discards and total catch.

Year	Official landings	IC landings	IC discards	IC total catch	Discard rate
2002	4414	4217	2084	6301	33.07%
2003	4110	3922	1370	5292	25.89%
2004	4772	4601	637	5238	12.16%
2005	4428	4214	1265	5479	23.09%
2006	5009	4837	1026	5863	17.50%
2007	4065	3908	2082	5990	34.76%
2008	3240	3067	1376	4443	30.97%
2009	3088	2804	1342	4146	32.37%
2010	3365	3166	3087	6253	49.37%
2011	3193	3041	1694	4735	35.77%
2012	2310	2189	1205	3394	35.50%
2013	1876	1750	1415	3165	44.71%
2014	2067	1907	1127	3034	37.15%
2015	1913	1762	1228	2990	41.07%
2016	1738	1750	628	2378	26.41%
2017	1344	1244	588	1832	32.10%
2018	1590	1587	657	2244	29.28%
2019	1670	1653	727	2380	30.55%

Year	Official landings	IC landings	IC discards	IC total catch	Discard rate
2020	1773	1715	679	2395	28.37%
2021	1328	1293	543	1836	29.58%
2022	1075	1046	257	1303	19.70%
2023	727*	714	346	1060	32.64%
2024	820*	797	503	1300	38.69%

*preliminary catch statistics

Table 6.8. Flounder in Subarea 4 and Division 3.a. Assessment summary. Weights are in tonnes. Catch, biomass index, harvest rate (Catch/biomass index) and length-based fishing pressure proxy (inverse $f = L_{F=M}/L_{mean}$) are given for the years used in the application of the *chr* rule. L_{mean} refers to the mean length above length at first capture and $L_{F=M}$ refers to the target reference length.

Year	Landings*	Discards	Catch	Stock size indicator (IBTSQ1 Index)	Harvest rate	Length-based fishing pressure proxy** ($L_{F=M}/L_{mean}$)
1950	5848					
1951	5143					
1952	4548					
1953	5096					
1954	4215					
1955	4753					
1956	4234					
1957	3521					
1958	3433					
1959	2962					
1960	3477					
1961	3179					
1962	3083					
1963	1866					
1964	2916					
1965	3340					
1966	2880					
1967	2003					

Year	Landings*	Discards	Catch	Stock size indicator (IBTSQ1 Index)	Harvest rate	Length-based fishing pressure proxy** ($L_F = M/L_{mean}$)
1968	2135					
1969	1709					
1970	1630					
1971	1897					
1972	1913					
1973	2348					
1974	5448					
1975	4406					
1976	4178					
1977	3624					
1978	3859					
1979	3396					
1980	2259					
1981	4153					
1982	4000					
1983	5581			32.345		
1984	3776			31.293		
1985	2098			21.833		
1986	2565			30.108		
1987	1923			48.932		
1988	2911			42.878		
1989	3080			51.558		
1990	1589			38.497		
1991	2283			49.778		
1992	1783			40.551		
1993	3288			37.387		
1994	3006			29.752		

Year	Landings*	Discards	Catch	Stock size indicator (IBTSQ1 Index)	Harvest rate	Length-based fishing pressure proxy** ($L_F = M/L_{mean}$)
1995	2623			25.916		
1996	2547			25.509		
1997	1727			28.529		
1998	6285			29.31		
1999	4260			23.23		
2000	3821			19.493		
2001	3727			17.379		
2002	4217	2084	6301	18.881		
2003	3922	1370	5292	21.196		
2004	4601	637	5238	24.225		
2005	4214	1265	5479	31.208		
2006	4837	1026	5863	28.225		
2007	3908	2082	5990	32.574		
2008	3067	1376	4443	36.358		
2009	2804	1342	4146	30.225		
2010	3166	3087	6253	26.513		
2011	3041	1694	4735	25.722		
2012	2189	1205	3394	26.226		
2013	1750	1415	3165	21.54		
2014	1907	1127	3034	22.144	137.04	0.85
2015	1762	1228	2990	30.906	96.73	0.95
2016	1750	628	2378	24.086	98.70	0.88
2017	1244	588	1832	24.811	73.85	0.82
2018	1587	657	2244	17.675	127.00	0.97
2019	1653	727	2380	18.819	126.45	0.91
2020	1715	679	2394	16.989	140.95	0.98
2021	1293	543	1836	18.656	98.40	0.98

Year	Landings*	Discards	Catch	Stock size indicator (IBTSQ1 Index)	Harvest rate	Length-based fishing pressure proxy** ($L_F = M/L_{mean}$)
2022	1046	257	1303	16.032	81.25	0.98
2023	714	346	1060	19.014	55.74	0.97
2024	797	503	1300	13.912	93.45	0.98
2025				***19.590		

* Official landings statistics 1950 – 2001, ICES-estimated landings 2002-2024.

** Only harvest rates in years where f ratio is above 1 are included in the calculation of $F_{MSY proxy}$

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